SMART INDIA HACKATHON | Problem Statement: SIH-1608 / MoR-TR-01

AI-Powered Automated Block Planning for Indian Railways

Exact Constraint Programming Decision Support System & Operations Control Centre (OCC) Digital Twin

Ministry of Railways
Government of India
Corridor: Delhi–Agra (265 KM)
Status: **Production Ready**

0	100%	6.9 ms	Zero
TRAIN CLASHES	HEADWAY COMPLIANCE	CP-SAT SOLVE TIME	LICENSE COST (FOSS)

1. EXECUTIVE SUMMARY

Indian Railways operates the 4th largest rail transport network on Earth, running over **13,000 passenger trains** and **8,000 freight rakes** daily across **68,000+ route kilometers**. To prevent catastrophic rail fractures, catenary snaps, and signal failures, infrastructure requires continuous preventative maintenance.

Performing physical maintenance requires a **"Maintenance Block" (Track Possession)**—suspending all train movements on that track section. Today, this is done manually via **phone calls, paper registers, and static spreadsheets** across 68 divisions. Controllers face an impossible dilemma: granting blocks causes cascading train delays, while deferring blocks leads to track deterioration and derailment hazards.

Our Solution: An enterprise-grade decision support platform that combines exact mathematical optimization (**Google OR-Tools CP-SAT**) with a real-time Control Room Digital Twin and Generative AI Co-Pilot. It deterministically schedules congested corridor possessions in **under 10 milliseconds** with mathematical guarantees of zero conflicts.

2. THE REAL-WORLD RAILWAY PROBLEM

The traditional block planning workflow suffers from three systemic bottlenecks:

- **The Punctuality vs. Infrastructure Dilemma:** Section controllers are held accountable to strict on-time punctuality metrics. Halting a prestigious train (e.g., *Rajdhani* or *Vande Bharat*) causes severe downstream junction congestion. Consequently, civil and electrical maintenance requests are repeatedly deferred. The CAG Report on Derailments in Indian Railways (Report No. 22 of 2022) cited track maintenance backlogs as a primary cause of rail fractures.
- **Cascading "Domino Effect" Delays:** A single poorly scheduled 90-minute track block on a double-line trunk route ripples across 500 kilometers, causing hours of unintended headway delays for following trains.
- **NP-Hard Combinatorial Complexity:** Scheduling 10 maintenance demands across 5 track sections with 15 scheduled trains, 2 maintenance gang crews, and interlocking constraints generates **millions of permutations**—surpassing human cognitive capacity.

3. SYSTEM ARCHITECTURE & SOLUTION LAYERS

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│                       OCC DIGITAL TWIN (FRONTEND)                           │
│  • Delhi-Agra 265 km corridor live topology (NDLS – GZB – ALG – TDK – MTJ) │
│  • 24-Hour Gantt timetable visualizer with real-time IST laser scrubber     │
│  • Live disruption and "what-if" chaos testing sandbox                      │
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                                        │ REST API (JSON)
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│                       HYBRID AI CO-PILOT LAYER                              │
│  • Field Voice & NLP Requisition Ingestion (English / Hindi / Hinglish)     │
│  • Explainable AI (XAI) transparent reasoning engine                        │
│  • Automated bilingual Indian Railways Form T/409 operating notice generator│
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                                        │ Mathematical Formulation
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│                 EXACT CP-SAT OPTIMIZATION ENGINE (BACKEND)                  │
│  • Google OR-Tools CP-SAT Constraint Programming Solver                     │
│  • Disjunctive non-overlapping interval constraints (AddNoOverlap)          │
│  • ±15-Minute train headway safety buffers                                  │
│  • Cumulative gangmen and crane capacity limits (AddCumulative)             │
│  • Sub-10ms deterministic execution (CPU-only, no GPUs required)            │
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4. MATHEMATICAL FORMULATION (CONSTRAINT PROGRAMMING)

Unlike generative models that can hallucinate, railway operations demand **100% mathematical certainty**. We formulate maintenance planning using Constraint Programming over Boolean Satisfiability (CP-SAT):

CONSTRAINT	MATHEMATICAL LOGIC	OPERATIONAL MEANING
Disjunctive Track Exclusivity	`model.AddNoOverlap(intervals)`	A train and a maintenance gang can **never** occupy the same track section simultaneously.
Safety Headway Buffers	`start − 15m, end + 15m`	Enforces a mandatory **±15-minute gap** before and after each train movement for deceleration and signaling clearance.
Cumulative Crew Limit	`model.AddCumulative(intervals, demands, max_crews)`	Restricts concurrent active blocks to available divisional gangs (e.g., maximum 2 concurrent crews).
Time Window Validity	`earliest_start ≤ start ≤ latest_end − duration`	Guarantees work occurs strictly within permissible engineering shifts (e.g., night-time possessions).
Priority Objective Function	`Maximize Σ (Priority × P_i) − Σ Deviation`	Prioritizes safety-critical work (P5 emergency repairs, P4 renewals) while minimizing deviation from preferred engineering times.

💡 **Why CPU-Only Matters for Indian Railways:**
Unlike deep learning architectures that require ₹50+ Lakh GPU clusters, our CP-SAT solver runs efficiently on standard **CPU datacenter hardware** already deployed across RailTel and CRIS divisional headquarters.

5. EMPIRICAL BENCHMARK: MANUAL VS. CP-SAT

Evaluated on an identical congested corridor dataset representing a 24-hour window on the Delhi–Agra trunk line (11 trains, 10 block demands, 2 maintenance gangs):

OPERATIONAL METRIC	UNOPTIMIZED / MANUAL BASELINE	AUTOMATED CP-SAT ENGINE	OPERATIONAL BENEFIT
Train Headway Clashes	6 Conflicted Trains	0 Clashes	100% Conflict Elimination
Estimated Train Delay	145 Minutes	0 Minutes	145 Minutes Saved per Corridor
Crew Over-Allocations	1 Over-Capacity Violation	0 Violations (≤ 2 Crews)	Safe Labor Compliance
Solve Computation Time	30–45 Minutes manual effort	**6.9 Milliseconds**	Instant Real-Time Dispatch
Safety P4/P5 Blocks Scheduled	Inconsistent	100% Guaranteed	Critical Asset Safety Preserved

6. UNIQUE FEATURES & OPERATIONAL VALUE

- **Control Room Digital Twin:** Visualizes the 265 km corridor with real-time station nodes, signal aspects (Green, Amber, Red), and live train tracking with bidirectional Down/Up line telemetry.
- **Disruption & "What-If" Simulator:** Instant dynamic re-solving during unexpected emergencies:
  - *Winter Fog Delay:* Injects +45 min delay on Rajdhani Express → CP-SAT shifts downstream blocks instantly.
  - *Emergency Rail Fracture:* Injects immediate P5 possession → locks the section and recalculates safe corridors.
  - *OHE Catenary Trip:* Models 25kV power de-energization.
- **Bilingual Form T/409 Circular:** Automatically generates official Indian Railways operating circular notices in English and Hindi for Station Masters and Section Controllers with zero manual drafting.
- **Explainable AI (XAI):** Answers plain-English inquiries from controllers regarding why specific possessions were shifted or deferred.

7. TECHNOLOGY STACK & ENTERPRISE INTEGRATION

LAYER	TECHNOLOGY	LICENSE	INTEGRATION RATIONALE
Constraint Solver	Google OR-Tools CP-SAT (v9.x)	Apache 2.0 (Free)	Global Gold Medal winner in MiniZinc scheduling competitions (2018–2024).
Backend API	FastAPI (Python 3.12)	MIT License (Free)	Asynchronous, sub-millisecond REST endpoints with Pydantic v2 validation.
Frontend	React 18 + TypeScript + Tailwind	MIT License (Free)	Responsive OCC UI with Indian Railways Crimson (`#7B1113`) & neutral obsidian theme.
Mapping & Schematics	ReactFlow	MIT License (Free)	Node-and-edge interactive railway corridor schematic.
Enterprise Data Handshake	CRIS COA, FOIS, ICMS, RTIS	Government of India	Native JSON schemas mapping directly to existing Indian Railways operational software.